



United International University
Department of Computer Science and Engineering

Project Report

Course Title: System Analysis and Design Laboratory
Course Code : CSE 3412

Section : E

Submitted To:
Sidratul Tanzila Tasmi
Lecturer of United International University

Group Member:
Saba Islam Jasi (011221005)
Sourav Karmakar (011221387)
Junayed Jawad Ahmad (011221511)
Anantha Debnath (011221534)
Mst. Rifat Ara Rimi (011221549)

Project Name : Go Trip (Smart Travel & Itinerary Planner)

Proposed Proposal:

Go Trip is a next-generation travel planning application designed to simplify and enhance the travel experience. Leveraging modern technologies, it offers intelligent itinerary creation, real-time travel data, and seamless booking integration. The platform aims to empower travelers with personalized tools that align with their preferences, budget, and travel goals.

Motivation:

The motivation behind developing Go Trip stems from a personal passion for travel and a desire to solve common pain points faced by modern travelers. Current travel planning tools are often fragmented, lacking personalization, real-time updates, and integration. By addressing these gaps, Go Trip aims to create a unified platform that simplifies travel planning, enhances user experience, and meets the evolving needs of tech-savvy travelers. The real-world relevance and potential to impact millions of travelers make this project both timely and meaningful.

Goals and Objectives:

- **Enhance User Experience:** Provide a user-friendly interface that simplifies the travel planning process.
- **Offer Personalization:** Deliver tailored recommendations based on individual preferences and travel history.

- **Ensure Convenience:** Integrate various services, including transportation, accommodation, and activity bookings, into a single platform.
- **Provide Real-Time Assistance:** Offer features like human translators, tour guide bookings, and emergency services to support travelers during their trips.

Scope:

This travel app is designed as a medium-scale mobile and web-based platform to assist users in planning and managing trips.

Included Features:

- User login and registration
- Bus and hotel booking
- Itinerary creation
- Notifications and booking history

Excluded Features:

- International payment gateway integration (not for now)
- In-app messaging or live chat
- AR/VR-based virtual tours
- Social media login integration

The app is mainly targeted for individual travelers and small travel groups, and can later be scaled for broader use.

Benchmark Analysis:

Feature	Triplt	Google Travel	TripAdvisor	Kayak	HolidayBD	GoZayaan	Our Project
1. Itinerary Creation							
2. Real-time Flight & Hotel Data							
3. Hotel & Flight Booking							
4. Personalized Recommendations							
5. Expense Tracking							
6. Social Sharing & Collaboration							
7. Offline Access							
8. Map Navigation Integration							
9. Book Tour Guide							
10. Nearby Garage For Car Service							

Feasibility Analysis:

Technical Feasibility:

- **Tools & Technology Available:** Modern tech stacks like React Native (for cross-platform app), Google Maps API, flight/hotel APIs (e.g., Skyscanner, Amadeus) are available and compatible.

- **Development Skills:** The team possesses the required skills in mobile app development, API integration, UI/UX, and cloud services.
- **Infrastructure:** Cloud platforms (e.g., Firebase, AWS) can support backend, authentication, and data storage.

Operational Feasibility:

- **User-Friendly Design:** Interface will be intuitive for travelers with various tech skills.
- **Maintenance:** App can be updated and maintained easily with modular development.
- **Availability of Real-Time Data:** APIs will ensure reliable real-time information access.

Economic Feasibility:

- **Low Initial Investment:** As an MVP (Minimum Viable Product), development costs are manageable with open-source tools and free-tier cloud services.
- **Future Monetization:** Potential revenue through ads, affiliate bookings, and premium features.
- **Cost-Benefit Analysis:** Expected benefits (convenience, integration, user satisfaction) outweigh the development and operational costs.

SWOT Analysis:

Strengths:

- All-in-one platform (itinerary, booking, expense tracking)
- User-friendly and personalized experience

- Real-time updates (flights, weather, events)
- Collaboration features

Weaknesses:

- Depends on third-party APIs for real-time data
- Initial version may lack offline access
- Requires internet and device compatibility

Opportunities:

- Growing demand for smart travel app
- Potential partnerships with airlines, hotels, and agencies
- Scope for premium features and AI-based travel planning

Threats:

- Competition from existing travel apps (e.g., Google Travel, Triplt)
- API limitations or pricing changes
- Security concerns with personal travel data

Feature List(Functional):

User Registration & Login

- Allows new users to sign up and returning users to log in securely.
- Includes password reset and social media login options

User Profile Management

- Users can update personal details, preferences, travel history, and saved itineraries.

Search Functionality

- Users can search for flights, hotels, destinations, tour guides, car services, etc.
- Includes filters by price, rating, date, location, etc.

Itinerary Creation

- Users can create, customize, and save travel itineraries including flights, hotels, activities, and notes.

Flight & Hotel Booking

- Real-time booking of flights and hotels from integrated APIs or databases.
- Includes date selection, pricing, availability, and confirmation.

Expense Tracking

- Users can add and track travel expenses, categorize them, and view budget summaries.

Social Sharing & Collaboration

- Users can share itineraries, photos, and reviews with friends or on social media.
- Collaborate with friends/family on shared trips.

Reviews & Ratings

- Users can leave feedback for hotels, guides, restaurants, and destinations.
- View average ratings and read reviews before booking.

Admin Panel

- Admin can manage user accounts, bookings, feedback, service listings, and promotion

Feature List (Non-Functional) :

Performance

- Fast load times and smooth navigation to enhance user engagement.

Usability

- User-friendly UI with an intuitive design and Available in multiple languages.

Reliability

- System should be available 99.9% of the time.

- Automatically recovers from crashes and prevents data loss during booking.

Scalability

- Ability to handle a growing number of users and data without performance degradation.

Maintainability

- Modular and well-documented codebase for easy updates and bug fixes.
- Future upgrades (new features or 3rd-party API changes) should be manageable.

Security requirements

- Secure login with encrypted password storage.
- Protects user data and payment information using HTTPS, SSL, and secure APIs.
- Includes role-based access control

Testing Techniques Implemented

Unit Testing

- Individual components like login, booking modules, and search filters were tested separately.
- This helped catch bugs at the function/method level during development.

Integration Testing

- After combining different modules (e.g., booking + payment, search + itinerary), we tested how they work together.
- This ensured smooth data flow and compatibility between components.

Black-box Testing

- We tested the app from the user's point of view without knowing the internal code.
- This helped validate expected outputs like successful logins, search results, and notifications.

User Acceptance Testing

- A group of users tested the app to see if it meets their expectations.
- Feedback was gathered on ease of use, clarity of features, and overall experience.

Technical Difficulties:

API Integration Issues

- Difficulty in integrating third-party APIs for real-time flight and hotel data due to limited access or usage restrictions.
- Some APIs required paid plans or complex authentication.

Compatibility Problems

- Ensuring the app worked smoothly across different browsers and devices (especially mobile vs desktop) was challenging.
- Responsive design issues had to be resolved.

Database Design Flaws

- Initial database structure did not scale well with multiple bookings and user types (e.g., travelers vs guides).
- Required redesign to support relationships and reduce redundancy.

Time Constraints

- Limited time led to prioritizing core features and leaving advanced features (e.g., AR, chatbot) for future versions.
- Testing and debugging under a short deadline was also a challenge.

Tool Limitations

- Some development tools lacked support for certain components or UI features.
- Switching tools or finding alternatives took extra time.

Conclusion:

In this project, we successfully designed and developed a feature-rich travel application aimed at simplifying the travel experience for users. Core functionalities such as user registration, searching and booking buses/hotels, itinerary creation, real-time location-based services etc. The system ensures a smooth and intuitive user experience, with a strong focus on usability, performance, and security. Users can manage their bookings, track expenses, get services - making the app a comprehensive travel companion.

Future Work :

In the future, we plan to enhance our travel app by focusing on the following key areas:

Offline Maps

We aim to integrate offline map functionality so users can access directions, locations, and travel routes even without an internet connection. This will be especially helpful for travelers in remote areas.

Tour Guide Integration

We plan to expand the tour guide feature by allowing users to book local guides, view guide profiles, ratings, and real-time availability, making the travel experience more informative and personalized.

Personalized Recommendations

We want to improve the recommendation system by using user preferences, past activities, and behavior to suggest destinations, hotels, restaurants, and experiences that match individual interests.

Nearby Services

The app will provide users with smart suggestions for nearby services such as car repair garages, ATMs, hospitals, restaurants, and emergency services based on their current location.

Notifications and Alerts

We will enhance the notification system to send real-time alerts about booking confirmations, cancellations, weather updates, price drops, upcoming trip reminders, and local travel advisories.

These features will significantly improve the app's usability, user satisfaction, and real-world travel support, making it more helpful and reliable for all types of travelers.